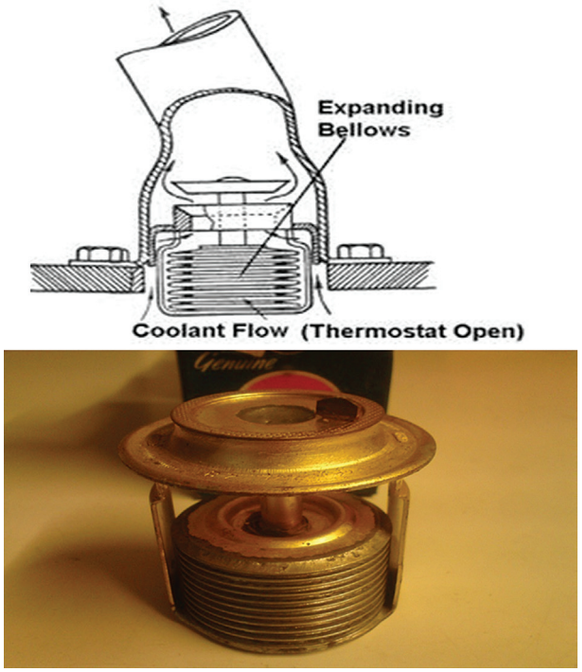


Figure 7.5.1 Bellows Type Thermostat

temperature reaches 70 to 80 deg Celsius, the chemicals inside the bellows get evaporated and allow the bellows to get expanded. The valve connected to the bellows now open and the water from the engine is sent to the upper tank of the radiator. When the cooling water temperature is reduced to 80 deg Celsius the chemicals inside the bellows get cooled and shrink. Now the valve connected to the bellows gets closed by this shrinkage. By this way thermostat controls cooling water circulation. The figure 7.8 shows the arrangement of the bellows type thermostat.

Comparison of air cooled and water cooled system:

Air cooled system	Water cooled system
Air is a medium of cooling	Water is medium of cooling
Fins are used	Water jacket, water pump radiator, thermostat like components are used.
Construction is easy	Construction is difficult
Less space is required for installation.	Needs more space
Produces more noise	Less noise
It runs with all the climatic changes.	In cold climatic countries water will freeze. So this type is not used in such countries.
No sediments or corrosion in the system	Corrosion and sedimentation occur.
Cooling process is not affected by any damage of one or two fins	If there is a water leak, it affects the cooling process.
This system cools the engine randomly.	This system cools the engine uniformly.
Conductivity of temperature is low	High
Not suitable for multi cylinder engines	Suitable for multi cylinder engines
Production and maintenance cost are low	high
Mostly used in two wheelers	Used for Lower, medium and high duty vehicles.

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